CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE

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College Code: 8226

Team ID: SWTID-2026-7898

TEAM DETAILS

Team Leader:

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Team Member 2:

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Team Member 3:

Name: Ravali R

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User Story:

The Customer Support Ticket Priority Prediction and Automated Assignment System uses Salesforce and Agentforce to analyze customer support ticket descriptions, automatically determine priority as High, Medium, or Low, and trigger backend automation for appropriate support handling. The solution uses the Support Ticket Intelligence custom object, an Auto-Launched Flow, and an Agentforce subagent named Support Ticket Priority Analysis. High-priority tickets create an urgent handling task and are assigned to a senior support agent, reducing manual effort and improving response time.

1. INTRODUCTION

1.1 Project Overview :

Customer support teams receive a large number of tickets daily, while prioritization and assignment can be manual. This project provides a Salesforce-based solution that analyzes ticket information, classifies urgency, assigns the appropriate support level, and initiates automation for critical cases.

1.2 Purpose :

- Automatically classify support tickets as High, Medium, or Low priority.
- Retrieve the latest support ticket associated with a customer Account.
- Reduce manual prioritization and assignment effort.
- Create an urgent handling task for High-priority tickets.
- Use Agentforce to analyze ticket details and trigger backend automation.
- Improve ticket response and resolution efficiency

2. IDEATION PHASE

2.1 Problem Statement

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Support Agent	focus on critical tickets first	tickets are not always prioritized automatically	ticket prioritization can be manual	frustrated
PS-2 |	Customer	get urgent issues resolved quickly	critical issues may wait behind lower-	support teams handle large ticket volumes	concerned

			priority tickets		
PS-3	Support Manager	assign tickets to the right support level	assignment may require manual decisions	ticket urgency is not consistently analyzed	overwhelmed
PS-4	Support Agent	identify urgent issues from descriptions	important keywords can be missed	ticket descriptions contain different urgency signals	uncertain
PS-5	Support Manager	reduce SLA-related risk	older unresolved tickets may remain unnoticed	SLA risk needs additional checking	concerned

2.2 Empathy Map Canvas

User Persona: Support Agent

- **Think & Feel:** Wants critical issues identified quickly so urgent customer problems receive attention.
- **Hear:** Receives customer issue descriptions and support requests.
- **See:** Ticket status, issue type, description, priority level, assignment, and SLA risk.
- **Say & Do:** Reviews tickets, handles assigned cases, and focuses on urgent support issues.
- **Pain:** Manual prioritization, missed urgency signals, and delays in handling critical tickets
- **Gain:** Automatic priority classification, clearer assignment, urgent-task creation, and reduced manual effort.

2.3 Brainstorming

The team brainstormed different approaches to automate customer support ticket prioritization and assignment. The final selected solution is a Salesforce CRM-based system integrated with Agentforce to analyze support ticket descriptions, classify ticket priority, and trigger appropriate backend automation.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey

Ticket Creation → Account Identification → Ticket Retrieval → Description Analysis → Priority Classification → Agent Assignment → Urgent Task Creation → Support Handling → Resolution

3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Support ticket creation and management
FR-2	Account and Contact association
FR-3	Ticket description analysis
FR-4	Automatic priority classification
FR-5	High-priority task creation
FR-6	Support agent assignment
FR-7	SLA breach risk checking
FR-8	Agentforce conversational analysis

Non-Functional Requirements

NFR	Description
Usability	Clear Agentforce interaction and understandable priority messages
Reliability	Consistent priority classification based on configured conditions
Performance	Quick retrieval and analysis of the latest support ticket
Security	Controlled Salesforce access to customer and ticket information
Maintainability	Flow and Agentforce configuration can be updated
Scalability	Supports increasing support-ticket volumes through Salesforce automation

3.3 Data Flow Diagram

Flow:

Customer / Support Request → Account → Latest Support Ticket → Ticket Description → Analyze Description → Priority Level → High/Medium/Low Decision → Agent Assignment → High-Priority Task (if applicable) → Final Response

Agentforce Flow:

User provides Account Name → Agentforce Support Ticket Priority Analysis → Auto-Launched Flow → Ticket Analysis → Priority and Assignment Output → Response to User.

3.4 Technology Stack

Layer	Technology
Platform	Salesforce
Database	Support_Ticket_Intelligence__c with Account, Contact, User and Task records
Logic	Auto-Launched Flow
AI	Agentforce
Agent Configuration	Support Ticket Priority Analysis subagent
Automation	Flow Decision, Assignment and Create Records
Output	Agentforce conversation response

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

Problem	Solution	Benefit
Manual ticket prioritization	Flow analyzes description and sets priority	Faster identification of urgent cases
Urgent issues may be missed	High-priority keyword decision	Critical cases receive attention
Manual task creation	Flow creates Task for High priority	Reduced manual effort
Manual support assignment	Flow sets assigned support level	More consistent handling
SLA risk can be overlooked	Optional SLA Risk decision	Better risk awareness
Need for conversational access	Agentforce subagent	AI-assisted interaction

4.2 Proposed Solution

Functional Requirements

Parameter	Description
Problem	Manual ticket prioritization and assignment causing delays in handling critical customer support issues
Solution	Salesforce-based Customer Support Ticket Priority Prediction and Automated Assignment System
Innovation	Agentforce AI + Auto-Launched Flow for analyzing ticket descriptions, automatically classifying priority, assigning support level, and triggering urgent-task automation
Impact	Faster identification of critical tickets, reduced manual effort, improved response time, and better support-team productivity
Business Model	Customer support operations involving customers, support agents, support managers, and AI-assisted ticket handling
Scalability	Supports increasing support-ticket volumes through Salesforce automation, with scope for additional assignment rules, SLA monitoring, and priority indicators

4.3 Solution Architecture

Components:

Users / Support Team → Agentforce → Support Ticket Priority Analysis → Auto-Launched Flow → Account & Ticket Data → Description Analysis → Priority Decision → Task / Assignment → Final Message → Agentforce Response

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

Product Backlog, Sprint Schedule, and Estimation

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story	Team Member	User Story / Task	Story Points	Priority
Sprint-1	Developer Setup	USN-1	**Kowsalya T**	As a system administrator, I want to configure the Salesforce environment for Support Ticket Intelligence so that the support automation solution can be developed.	3	High
Sprint-2	Data Modeling	USN-2	**Deepika M**	As an admin, I want to create the Support Ticket Intelligence object with required fields and relationships so that customer support ticket information is stored systematically.	5	High
Sprint-2	Data Modeling	USN-3	**Sukitha M**	As a support user, I want support tickets to be associated with Accounts and Contacts so that the latest customer ticket can be identified for analysis.	3	High
Sprint-3	Automation	USN-4	**Ravali R**	As a support agent, I want an Auto-Launched Flow to retrieve the latest ticket and analyze its description so that ticket priority can	5	High

				be determined automatically.		
Sprint-3	Automation	USN-5	**Ravali R**	As a support agent, I want tickets to be automatically classified as High, Medium, or Low based on configured urgency conditions so that critical issues receive attention first.	5	High
Sprint-4	Agentforce & Automation	USN-6	**Kowsalya T**	As an AI Agent (Agentforce), I want to analyze support ticket details and trigger the backend Flow so that manual ticket prioritization is reduced.	8	High
Sprint-4	Task & Assignment	USN-7	**Deepika M**	As a support manager, I want High-priority tickets to automatically create an urgent handling task and assign the appropriate support level so that critical tickets are handled quickly.	5	High
Sprint-5	SLA Management	USN-8	**Sukitha M**	As a support manager, I want the system to check SLA breach risk for older unresolved tickets so that potential SLA issues can be identified.	5	Medium

Sprint-6	Agentforce & Conversational Support	USN-9	**Ravali R**	As a support user, I want to provide an Account Name to Agentforce and receive the ticket ID, priority, assigned agent, and action message so that I can understand the ticket handling outcome conversationally.	5	Medium

6. Project Development Phase:

Milestone 1: Salesforce Account

Activity 1: Creating Developer Account

Creating a developer org in Salesforce.

1. Go to Salesforce Developer signup page.
2. On the sign up form, enter the following details:
    1. First Name & Last Name
    2. Email
    3. Role: Developer
    4. Company: College Name
    5. Country: India
    6. Postal Code: Pin Code
    7. Username: Should be a combination of your name and company

This need not be an actual email ID. You can give anything in the format:

username@organization.com

Click on Sign Me Up after filling these details.

Activity 2: Account Activation

1. Go to the inbox of the email that you used while signing up.

2. Click on **Verify Account** to activate your account. The email may take 5–10 minutes.

Milestone 2: Data Management-Objects

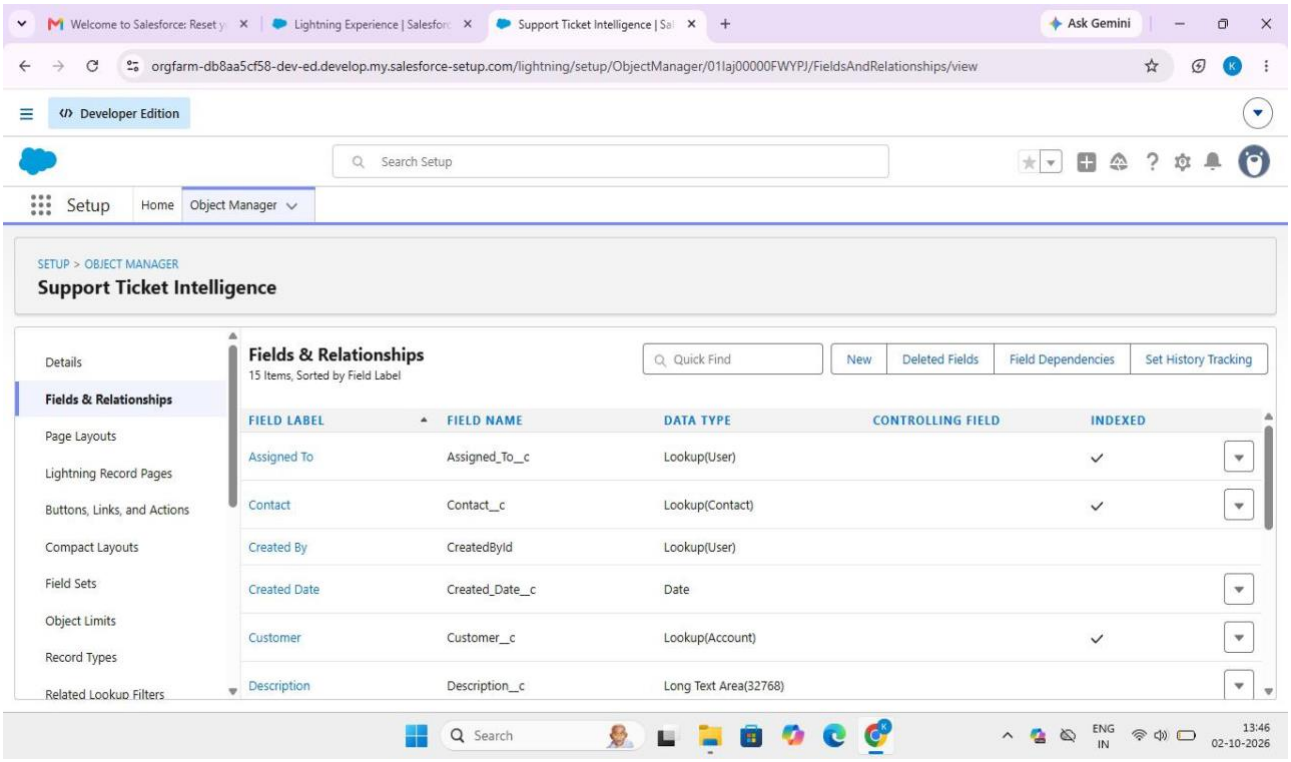
Custom Object

Object Name:
Support Ticket Intelligence

API Name:
Support_Ticket_Intelligence__c

Fields

Field Label	API Name	Data Type	Description
Ticket Number	Ticket_Number__c	Auto Number	TKT-{0000}
Customer	Customer__c	Lookup (Account)	Related Account
Contact	Contact__c	Lookup (Contact)	Customer contact
Issue Type	Issue_Type__c	Picklist	Technical, Billing, General
Description	Description__c	Long Text Area	Issue details
Priority Level	Priority_Level__c	Picklist	Low, Medium, High
Status	Status__c	Picklist	New, In Progress, Resolved
Created Date	Created_Date__c	Date	Ticket date
Assigned To	Assigned_To__c	Lookup (User)	Support agent
SLA Breach Risk	SLA_Breach_Risk__c	Checkbox	Risk flag
Resolution Time (hrs)	Resolution_Time__c	Number	Time taken



Auto-Launched Flow Logic (Modified for this Project)

1. Flow Type

Auto-Launched Flow (No Trigger or called from Agentforce)

2. Variables

Input Variable

Label	API Name	Type	Input
Account Name	varAccountName	Text	Yes

Output Variables

Label	API Name	Type	Output
Account Id	varAccountId	Text	Yes
Ticket Id	varTicketId	Text	Yes
Priority Level	varPriorityLevel	Text	Yes
Assigned Agent	varAssignedTo	Text	Yes
Action Message	varActionMessage	Text	Yes

3. Step-by-Step Flow Implementation

Step 1: Get Account

Element: Get Records

- Label: Get Account
- Object: Account
- Condition:
  Name Equals {!varAccountName}
- Sort: CreatedDate DESC
- Store: Only first record

Step 2: Assignment (Store Account Id)

Element: Assignment

- Label: Store Account Id

Set:

- varAccountId = {!Get_Account.Id}

Step 3: Get Ticket Record

Element: Get Records

- Label: Get Ticket
- Object: Support_Ticket_Intelligence__c
- Condition:
  Customer__c Equals {!varAccountId}
- Sort: CreatedDate DESC
- Store: Only first record

Step 4: Assignment (Store Ticket Id)

Element: Assignment

- Label: Store Ticket Id

Set:

- varTicketId = {!Get_Ticket.Id}

Step 5: Decision (Analyze Description)

Element: Decision

- Label: Analyze Description

Outcomes:

High Priority

Description contains:

- "urgent"
- "not working"
- "failure"

(Use multiple conditions with CONTAINS)

Medium Priority

Description contains:

- "issue"
- "slow"
- "delay"

Default Outcome

- Low Priority

Step 6: Assignment (Set Priority Level)

High Path

- varPriorityLevel = "High"

Medium Path

- varPriorityLevel = "Medium"

Low Path

- varPriorityLevel = "Low"

Step 7: Decision (High Priority Check)

Element: Decision

- Label: Is High Priority

Condition:

- varPriorityLevel Equals "High"

Step 8: Create Task (Only High Priority)

Element: Create Records

Object: Task

Set Fields:

- Subject = "Urgent Ticket Handling"
- WhatId = {!varTicketId}
- Priority = High
- Status = Not Started

Step 9: Assignment (Assign Agent Name)

Element: Assignment

- Label: Set Assigned Agent

Set:

- varAssignedTo = "Senior Support Agent"

Step 10: Optional SLA Check

Decision: SLA Risk

Condition example:

- If Created Date older than 2 days

Then:

- (Optional internal flag or message)

Step 11: Final Output Message

Element: Assignment

- Label: Set Final Message

Set varActionMessage:

High:
“High priority ticket detected. Assigned to senior agent.”

Medium:
“Ticket marked as medium priority. Will be handled shortly.”

Low:
“Tickets are low priority and queued for processing.”

Step 12: Save Flow

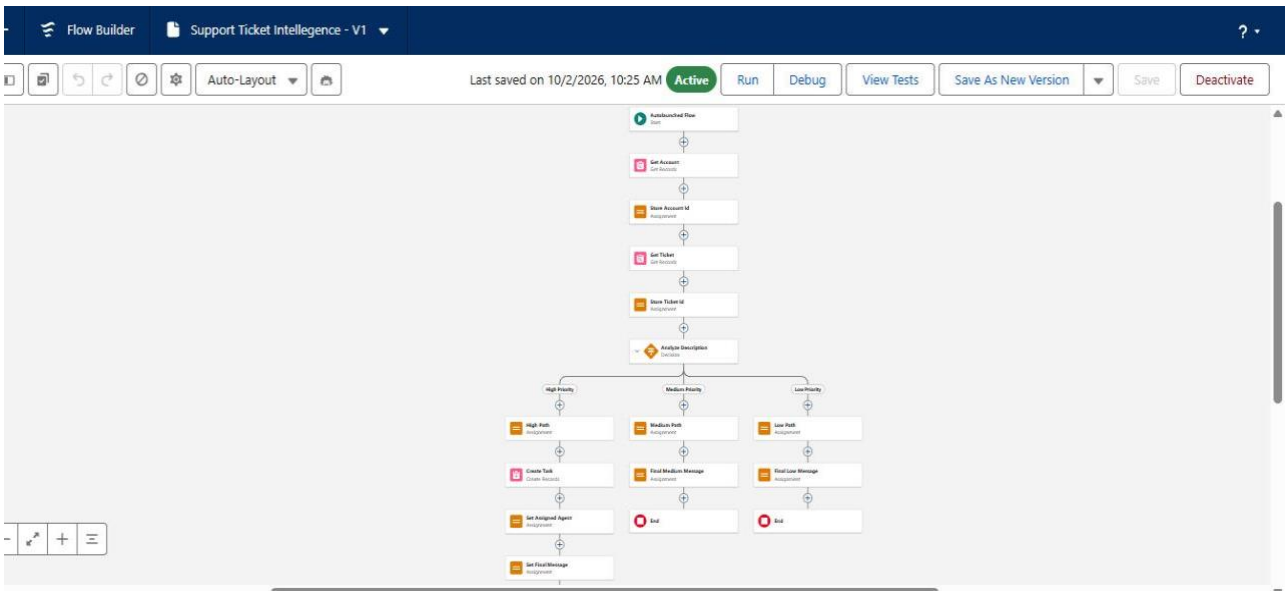
Label Name: Support_Ticket_Intelligence

Active Flow

4. Final Flow Structure

Start
→ Get Account
→ Assignment (Account Id)
→ Get Ticket
→ Assignment (Ticket Id)
→ Decision (Analyze Description)
→ Assignment (Priority Level)
→ Decision (High Priority)
→ Create Task
→ Assignment (Assigned Agent)
→ Optional SLA Decision
→ Assignment (Final Message)
→ End

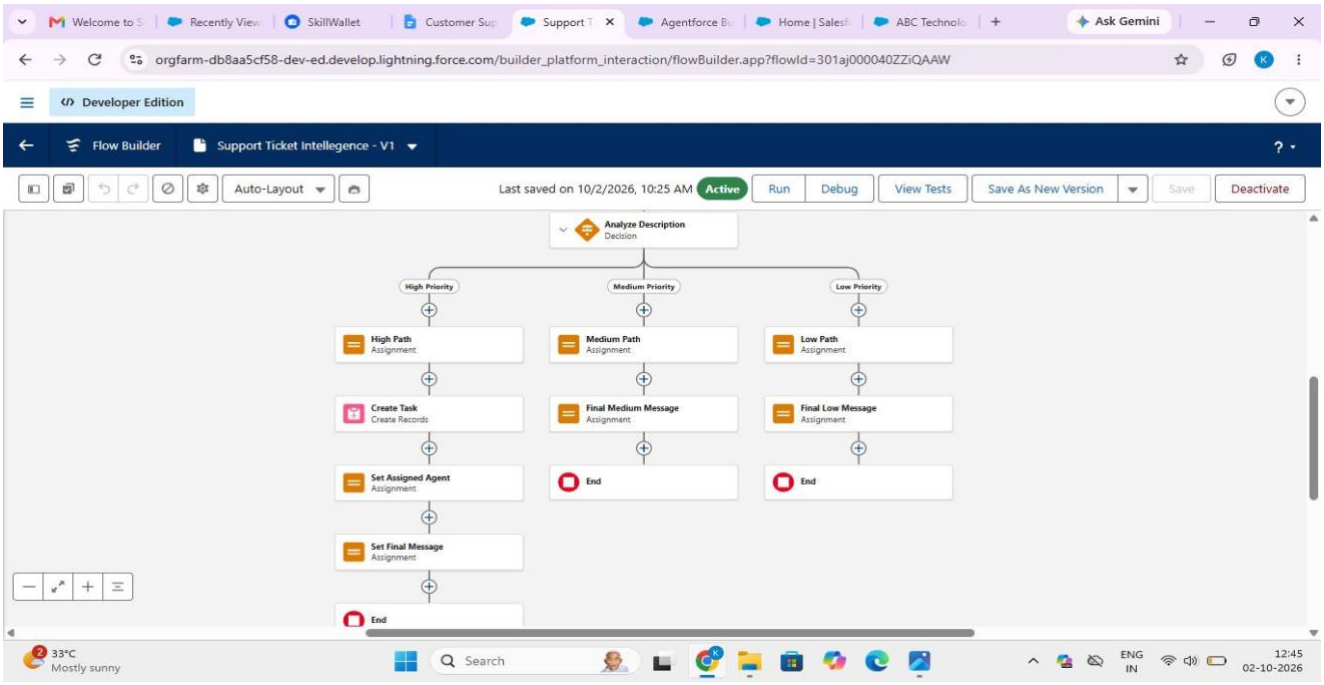
Figure: Support Ticket Intelligence Auto-Launched Flow



5. Flow Output

The flow retrieves the Account based on the provided Account Name, identifies the latest support ticket, analyzes the ticket description, determines the priority level, assigns the appropriate support level, creates an urgent task for High-priority tickets, and returns the final action message.

Figure: Successful Flow Debug Execution



Output includes:

- Account Id
- Ticket Id
- Priority Level
- Assigned Agent
- Action Message

6. Final Outcome

- Takes Account Name as input
- Fetches latest support ticket
- Analyzes description using keywords
- Sets priority automatically
- Creates task for urgent cases
- Returns output to Agentforce

Agentforce Configuration

Steps:

1. Go to Gear icon and Click Salesforce Go.
2. Search Default and Select Agentforce Default then Enabled It.
3. Select Get Started → Turn On → Confirm.
4. Search Agents in Quick Find Box and Enabled The Agent then you will see Default Agent.
5. Select **Default Agent** and Click **Open in Builder**.
6. Click **New** to Create New SubAgent.
7. Click **Next**.

Agentforce Subagent Configuration

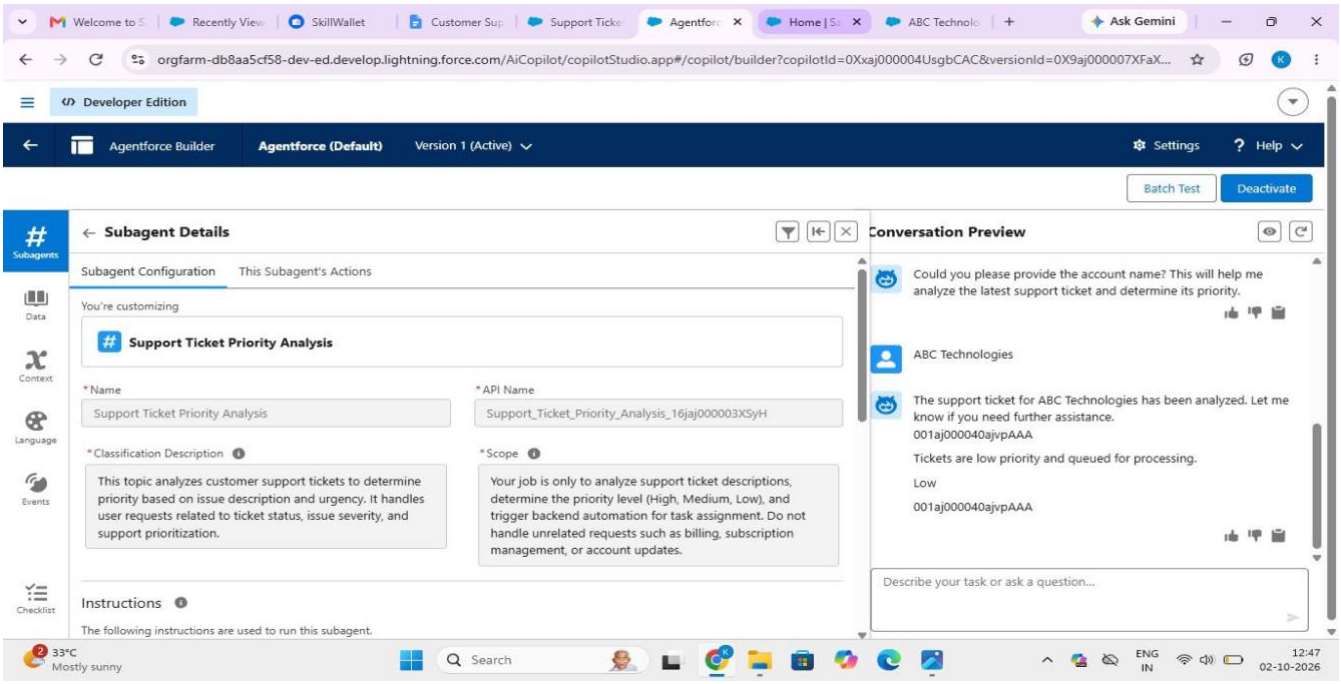
Name
Support Ticket Priority Analysis

API Name
Support_Ticket_Priority_Analysis

Classification Description
This topic analyzes customer support tickets to determine priority based on issue description and urgency. It handles user requests related to ticket status, issue severity, and support prioritization.

Scope
Your job is only to analyze support ticket descriptions, determine the priority level (High, Medium, Low), and trigger backend automation for task assignment. Do not handle unrelated requests such as billing, subscription management, or account updates.

Figure: Support Ticket Priority Analysis Subagent Configuration



Instructions

Follow these steps when this topic is triggered:

1. Ask the user to provide the **Account Name** if not already given.
2. Retrieve the latest support ticket associated with the Account.
3. Read and analyze the ticket description.
4. Determine priority based on keywords:

High Priority: contains “urgent”, “not working”, “failure”

Medium Priority: contains “issue”, “slow”, “delay”

Low Priority: if none of the above

5. Set the priority level accordingly.
6. If the ticket is **High Priority:**
    - Trigger the Auto-Launched Flow
    - Ensure a task is created for immediate handling
7. Assign the ticket to the appropriate support level (e.g., senior agent for high priority).

8. Return a clear response message to the user:
    - High → Inform urgent handling is initiated
    - Medium → Inform issue will be handled shortly
    - Low → Inform ticket is queued
9. Do not update records directly or send emails. Use Flow for all actions.

Agent Action Description

Analyzes customer support tickets based on the provided account name, determines ticket priority using issue description, and triggers automation to assign tasks for high-priority cases.

Loading Text

Analyzing ticket details and determining priority...

Input

varAccountName

Description:
Enter the customer account name to fetch the latest support ticket and analyze its priority.

Data Type:
lightning__textType

Require Input:
Enabled

Collect Data From User:
Enabled

Outputs

varAccountId

Description:
Returns the unique identifier of the customer account used to retrieve the support ticket.

Data Type:
lightning__textType

Show in Conversation:
Enabled

Output Rendering:
Text

varActionMessage

Description:
Displays the final message indicating ticket priority and actions taken by the system.

Data Type:
lightning__textType

Show in Conversation:
Enabled

Output Rendering:
Text

varAssignedTo

Description:
Indicates the support agent or team assigned to handle the ticket.

Data Type:
lightning__textType

Show in Conversation:
Enabled

Output Rendering:
Text

varPriorityLevel

Description:
Represents the calculated ticket priority level (High, Medium, Low) based on issue description.

Data Type:
lightning__textType

Show in Conversation:
Enabled

Output Rendering:
Text

varTicketId

Description:
Returns the unique identifier of the support ticket analyzed by the system.

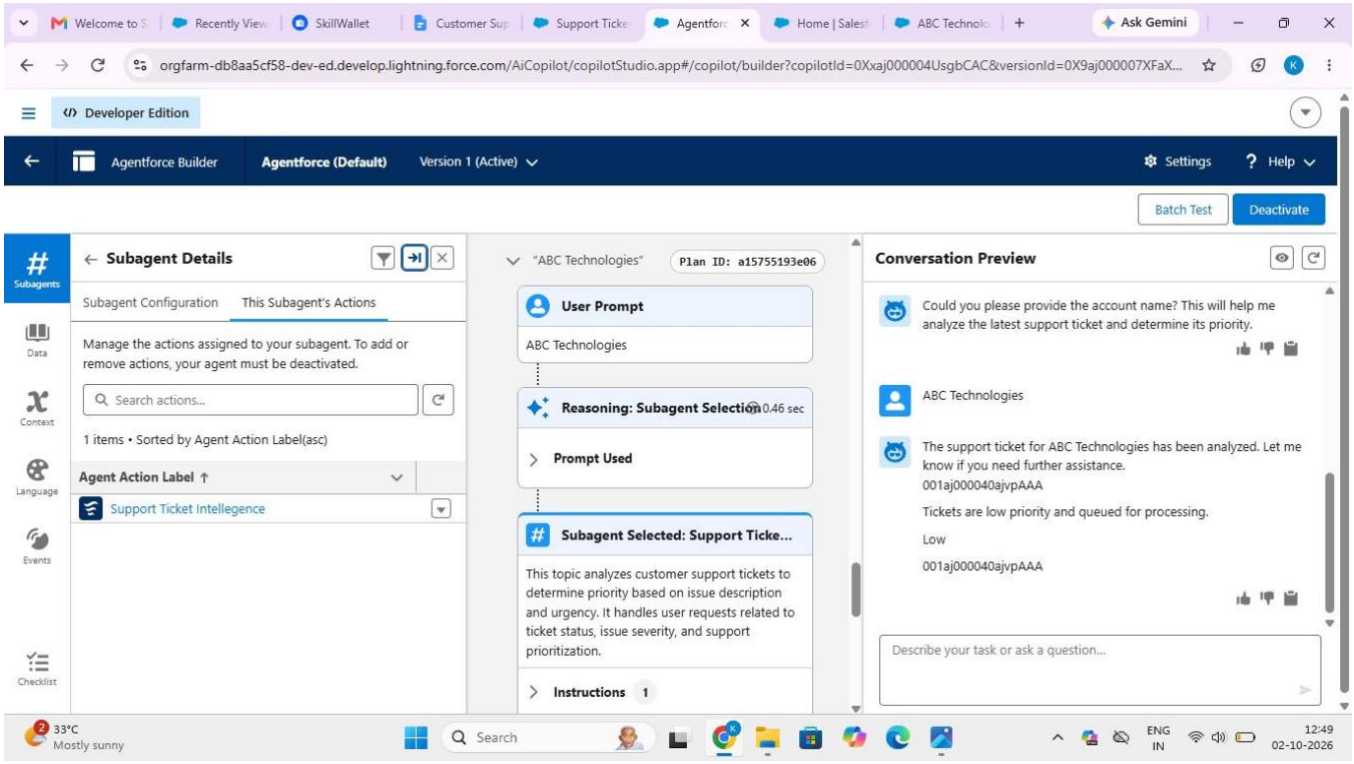
Data Type:
lightning__textType

Show in Conversation:
Enabled

Output Rendering:
Text

Final Output:

Figure: Agentforce Final Ticket Priority Analysis Output



7. FUNCTIONAL & PERFORMANCE TESTING

During testing, the Salesforce implementation was validated to ensure that the Customer Support Ticket Priority Prediction and Automated Assignment System functions correctly. The Auto-Launched Flow successfully retrieves the latest support ticket associated with the given

Account, analyzes the ticket description, and classifies the priority as High, Medium, or Low based on the configured conditions.

For High-priority tickets, the system successfully creates an Urgent Ticket Handling task and assigns the appropriate support level as Senior Support Agent. The Flow was also tested through Agentforce using the Support Ticket Priority Analysis subagent. Agentforce successfully triggered the backend Flow and returned the ticket priority, assigned agent, and action message in the conversation.

The successful Flow execution and Agentforce response confirm that the ticket prioritization, task creation, assignment, and conversational support features are functioning as expected.

Testing Result: Successful

Test Case 1: High Priority Ticket Analysis

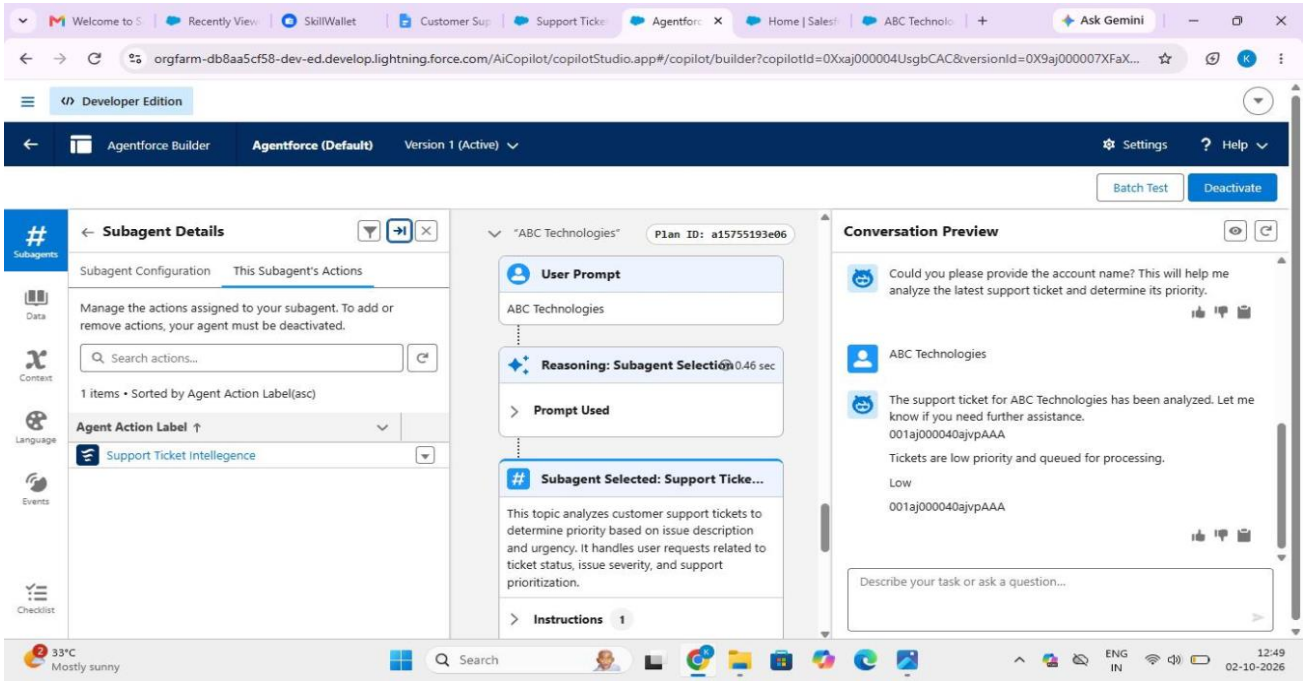
Expected Output:

The system should identify the ticket as High Priority, create an urgent handling task, assign it to the Senior Support Agent, and return the appropriate action message through Agentforce.

Actual Output:

The system successfully identified the ticket as High Priority and returned the message “High priority ticket detected. Assigned to senior agent.” The assigned support level was displayed as “Senior Support Agent.”

Result: Pass



Test Case 2: Auto-Launched Flow Execution

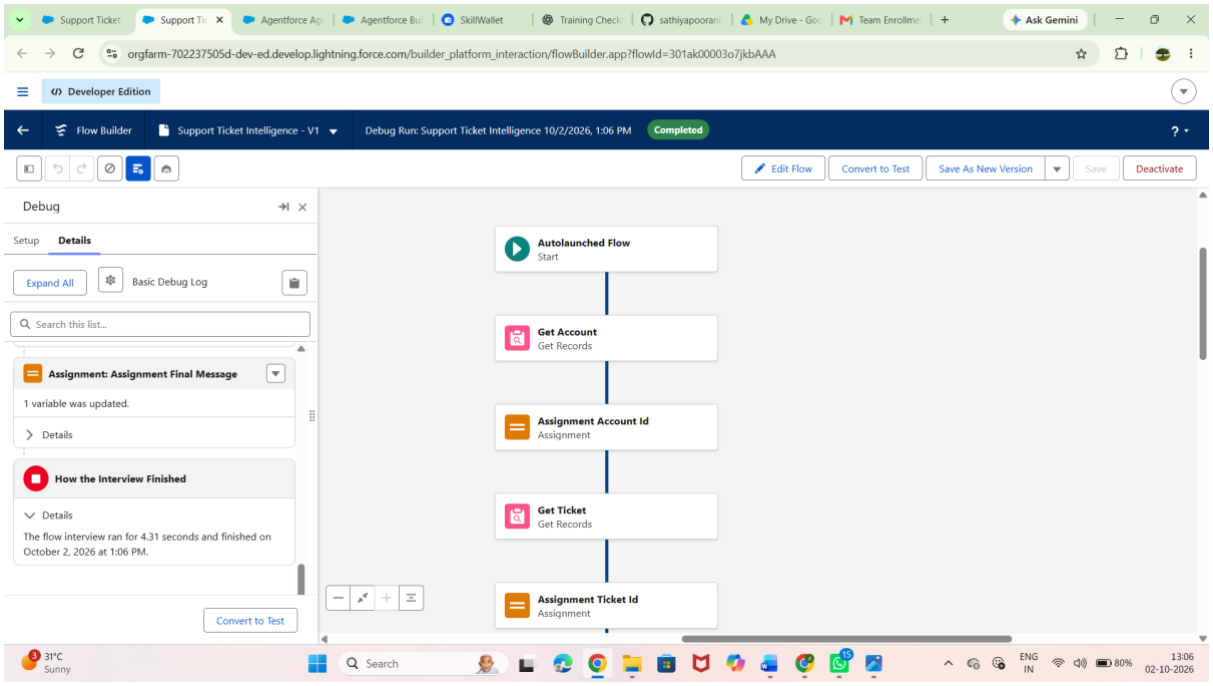
Expected Output:

The Auto-Launched Flow should retrieve the Account, fetch the latest associated support ticket, analyze the ticket description, determine the priority level, and return the required output values.

Actual Output:

The Auto-Launched Flow executed successfully. It retrieved the Account and support ticket, analyzed the description, determined the ticket priority, and completed the configured flow path successfully.

Result: Pass



Test Case 3: High-Priority Task Creation

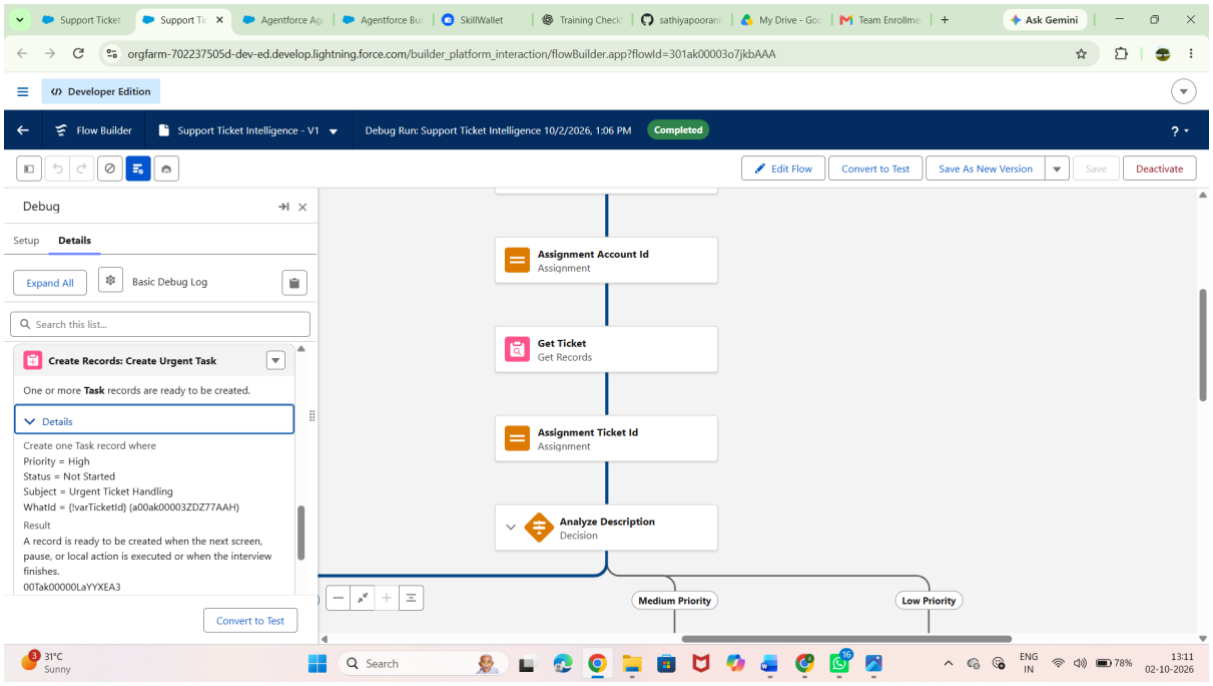
Expected Output:

When a support ticket is classified as High Priority, the Flow should create a Task with the subject “Urgent Ticket Handling”, Priority as High, and Status as Not Started.

Actual Output:

For the High Priority path, the Flow successfully executed the configured Create Records element for the urgent ticket handling process.

Result: Pass



Test Case 4: Agentforce and Flow Integration

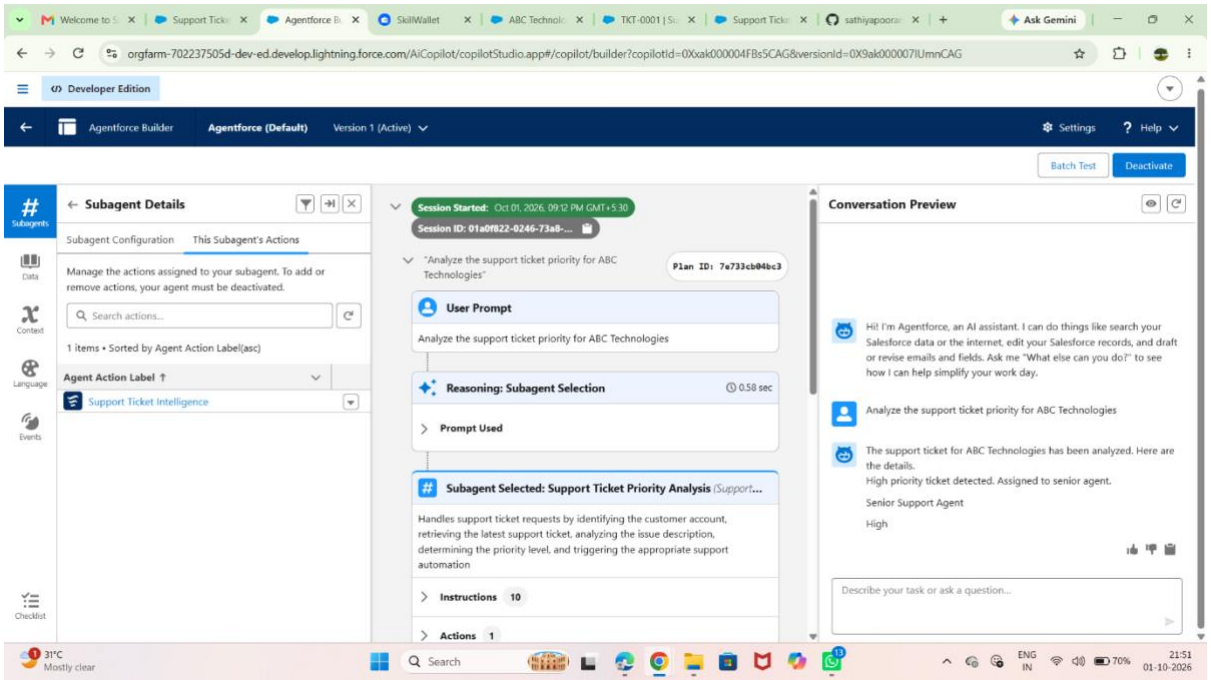
Expected Output:

Agentforce should accept the customer Account Name, trigger the Support Ticket Intelligence Flow, and return the ticket priority, assigned support agent, and action message.

Actual Output:

Agentforce successfully triggered the backend Flow for the provided Account Name and returned the High priority result, “Senior Support Agent”, and the corresponding action message.

Result: Pass



Test Case 5: Priority Classification Logic

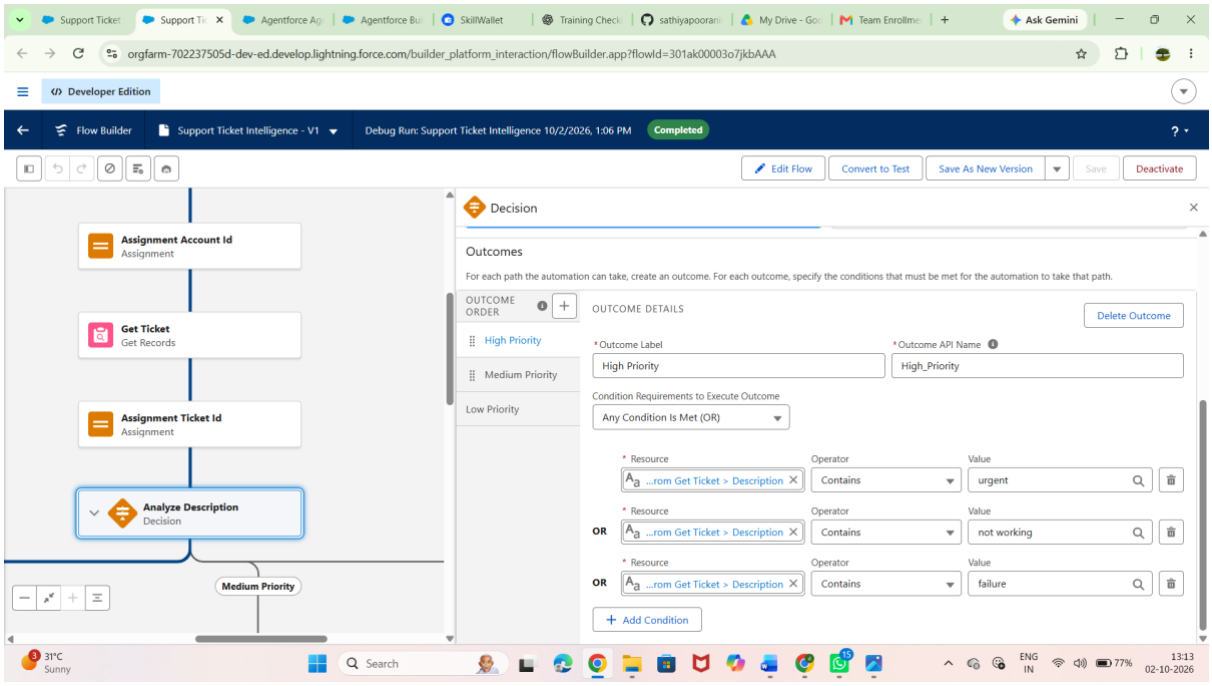
Expected Output:

The system should classify tickets containing keywords such as “urgent”, “not working”, or “failure” as High Priority.

Actual Output:

The Flow successfully analyzed the ticket description and classified the tested support ticket as High Priority based on the configured keyword conditions.

Result: Pass



8. ADVANTAGES & DISADVANTAGES

Advantages

- Automatic Ticket Prioritization – Classifies support tickets as High, Medium, or Low based on ticket description.
- Faster Identification of Critical Issues – Helps support agents focus on urgent tickets first.
- Reduced Manual Effort – Automates ticket analysis, priority setting, task creation, and assignment.
- Intelligent Ticket Handling – Agentforce analyzes ticket details and initiates the appropriate backend automation.
- Automatic Urgent Task Creation – High-priority tickets automatically create an Urgent Ticket Handling task.
- Improved Support Assignment – Assigns High-priority tickets to the appropriate support level, including a senior support agent.
- Conversational Support – Support users can interact with Agentforce to analyze ticket information and receive the outcome.
- Better Response Time – Automation helps reduce delays in handling critical customer issues.
- SLA Risk Awareness – The solution can check older unresolved tickets for potential SLA breach risk.

- Scalable Support Operations – Salesforce automation can support increasing volumes of support tickets.
- Improved Team Productivity – Reduces repetitive work and allows support teams to concentrate on ticket resolution.

Disadvantages

- Keyword-Based Priority Logic – The documented priority classification depends on configured keywords such as urgent, not working, failure, issue, slow, and delay.
- Limited Understanding of Context – The current implementation may not capture every possible urgency signal because the documented logic is based on configured conditions.
- Dependence on Accurate Ticket Data – Correct Account, Contact, and ticket information is required for successful analysis.
- Flow Configuration Dependency – Changes to ticket handling require maintenance of the Auto-Launched Flow and its logic.
- Agentforce Configuration Dependency – Agentforce responses and actions depend on properly configured subagent instructions and Flow actions.
- Limited Scope of Agentforce – The documented Agentforce subagent is focused on ticket analysis, priority determination, and backend automation; unrelated requests such as billing or subscription management are outside its scope.
- SLA Monitoring is Limited – The current implementation includes an optional SLA-risk check rather than a comprehensive SLA monitoring and escalation solution.
- Assignment Logic May Need Expansion – Additional support teams or more sophisticated assignment rules would require further configuration.
- Maintenance Required – Priority keywords, Flow variables, Agentforce outputs, and assignment logic may need to be updated as support requirements evolve.
- Limited Analytics in Current Scope – Advanced analytics for priority trends and resolution performance are identified as future scope rather than part of the current implementation.

9. Conclusion:

The Customer Support Ticket Priority Prediction and Automated Assignment System demonstrates how Salesforce Flow and Agentforce can work together to reduce manual support-ticket handling. The solution retrieves the latest ticket for an Account, analyzes its description, classifies priority, creates an urgent task for High-priority cases, assigns a support level, and returns a clear conversational response. This

supports faster ticket handling, reduced manual effort, intelligent prioritization, and better team productivity.

10. FUTURE SCOPE

- Add more priority indicators and business rules.
- Extend Agentforce outputs with additional ticket information.
- Add more sophisticated SLA monitoring and escalation.
- Support additional support teams or assignment rules.
- Introduce analytics for priority trends and resolution performance.
- Add more variables and configure them in the Agentforce action for richer outputs.

11. APPENDIX

- **A. Custom Object:** Support Ticket Intelligence
- **B. Key Salesforce Records:** Account, Contact, User, Task
- **C. Flow:** Support_Ticket_Intelligence - V1– Auto-Launched Flow
- **D. Agentforce Subagent:** Support Ticket Priority Analysis
- **E. Agentforce Action:** Flow-based Support Ticket Intelligence action

Thank You